

Genome Organization

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Intended Learning Outcomes

1. Define the biological architecture of a genome and distinguish between the structural organization found in prokaryotic and eukaryotic genetic systems.
2. Explain the mechanical function of topoisomerases and structural proteins in driving prokaryotic chromosome supercoiling.
3. Classify eukaryotic chromosomal architecture into coding sequences, regulatory motifs, and repetitive regions.
4. Illustrate the cellular roles of centromeres and telomeres during chromosomal division and protect linear DNA sequences from degradation.

Introduction

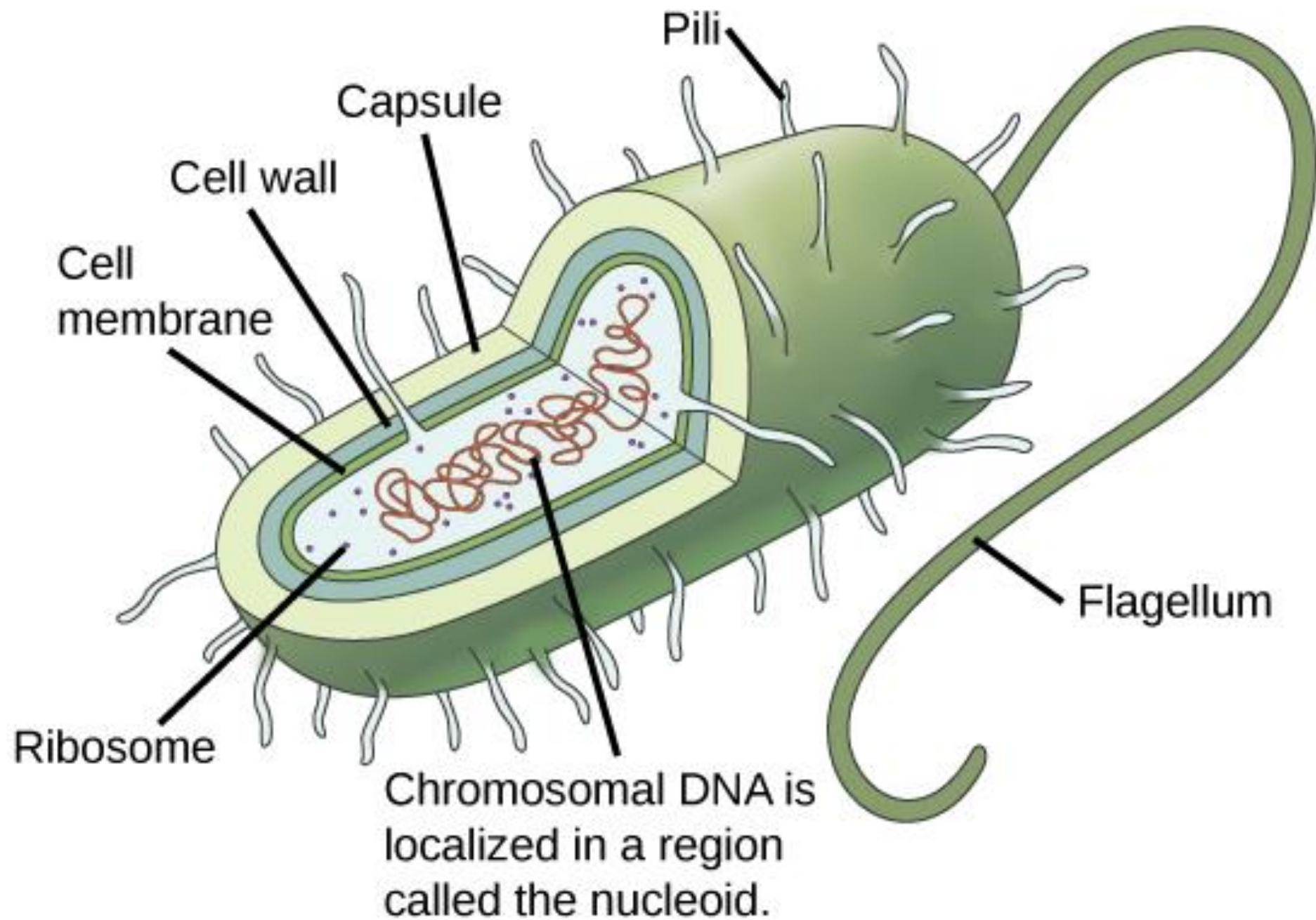
- The **genome** is the complete set of genetic instructions in an organism.
- While a sequence of bases may seem simple, a genome is a highly structured and organized entity, with a complex architecture that is fundamental to its function.
- The way a genome is organized differs dramatically between **prokaryotic** and **eukaryotic** organisms, reflecting their differences in cell structure and complexity.

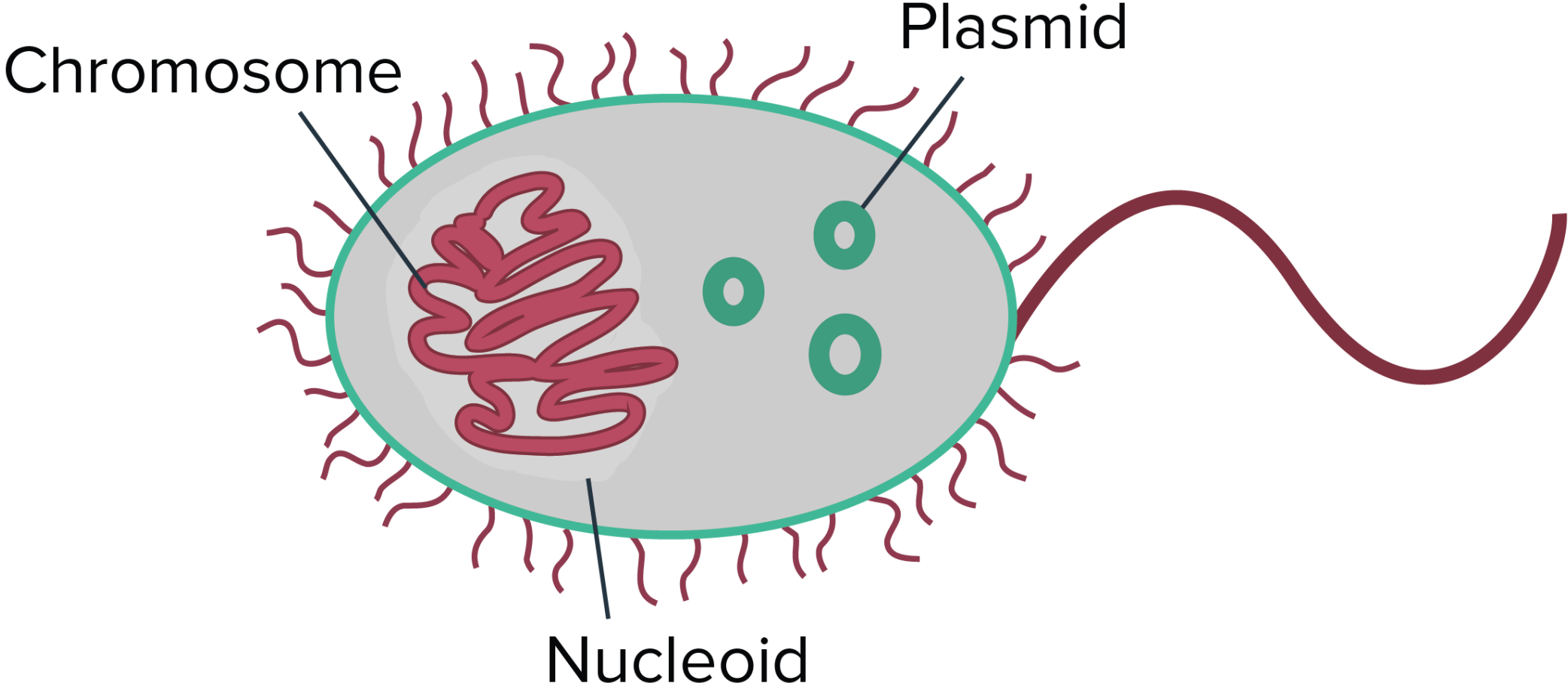
Prokaryotic Genome Organization

Prokaryotes (bacteria and archaea) have a relatively simple and compact genome.

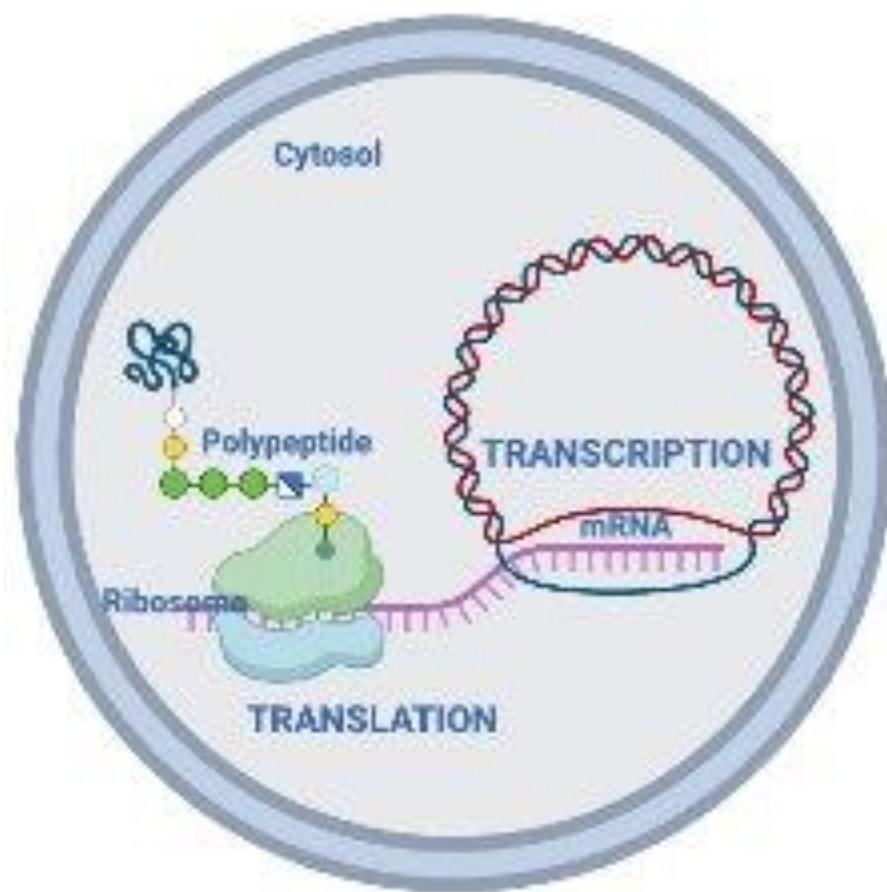
General Characteristics:

- **Single, Circular Chromosome:** The main genetic material is a single, large, circular DNA molecule.
- **Location:** This chromosome is located in a dense region of the cytoplasm called the **nucleoid**, as there is no membrane-bound nucleus.
- **Supercoiled:** To fit inside the cell, the circular chromosome is twisted and coiled upon itself in a process called **supercoiling**.

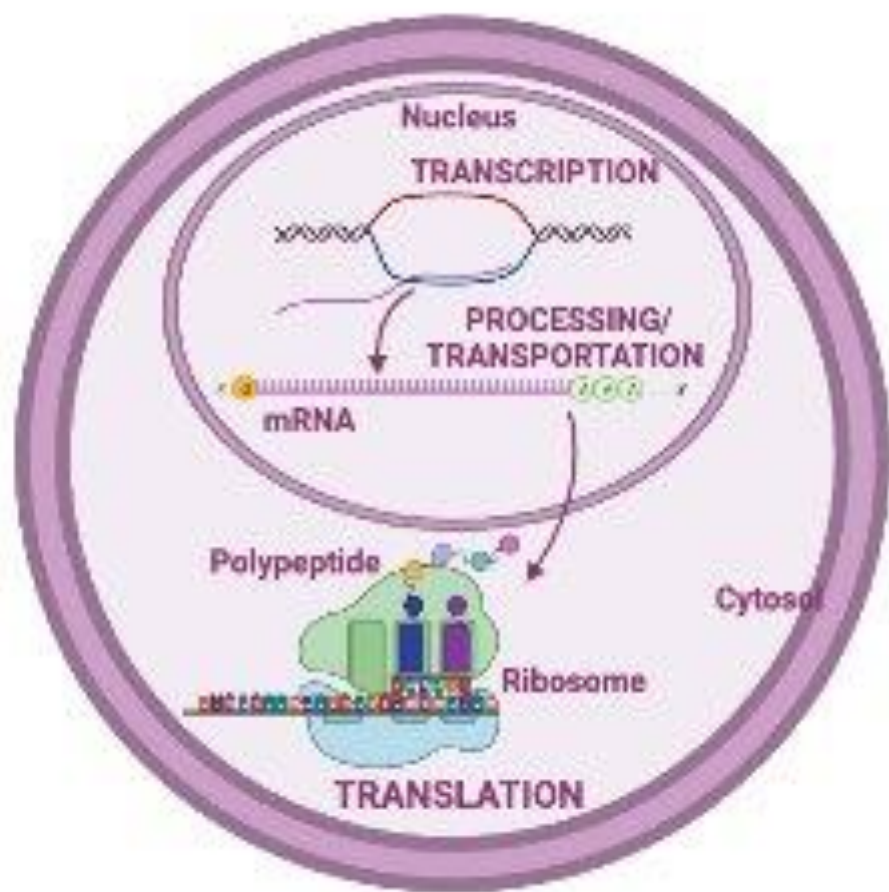




PROKARYOTIC CELL

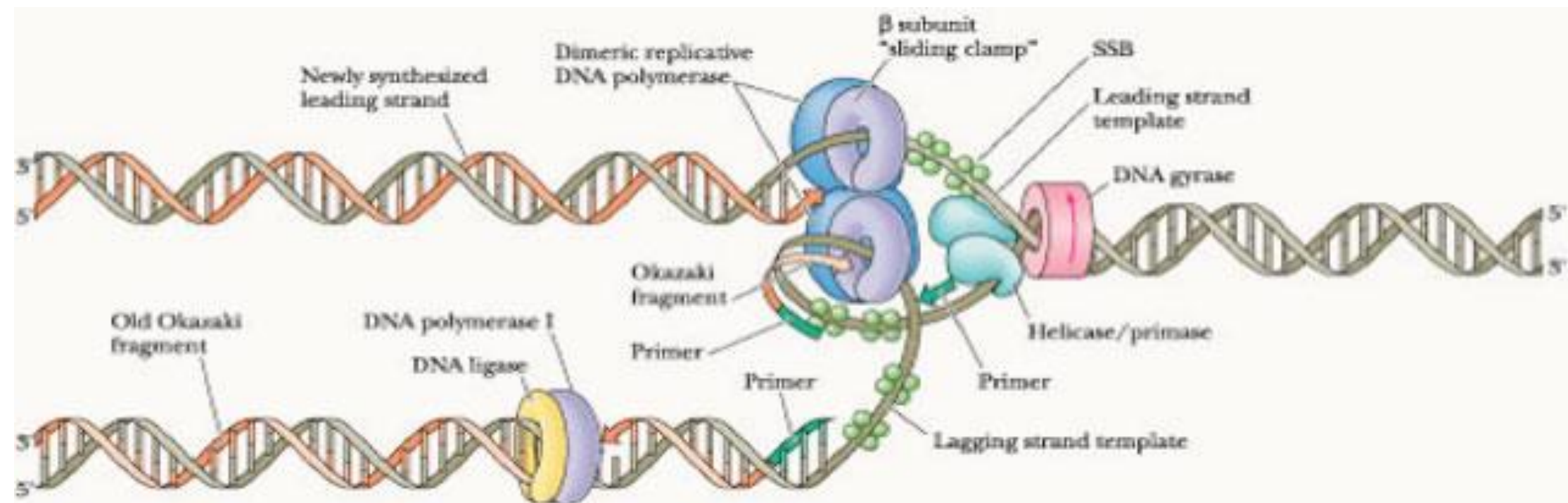


EUKARYOTIC CELL



Structure of the Prokaryotic Chromosome

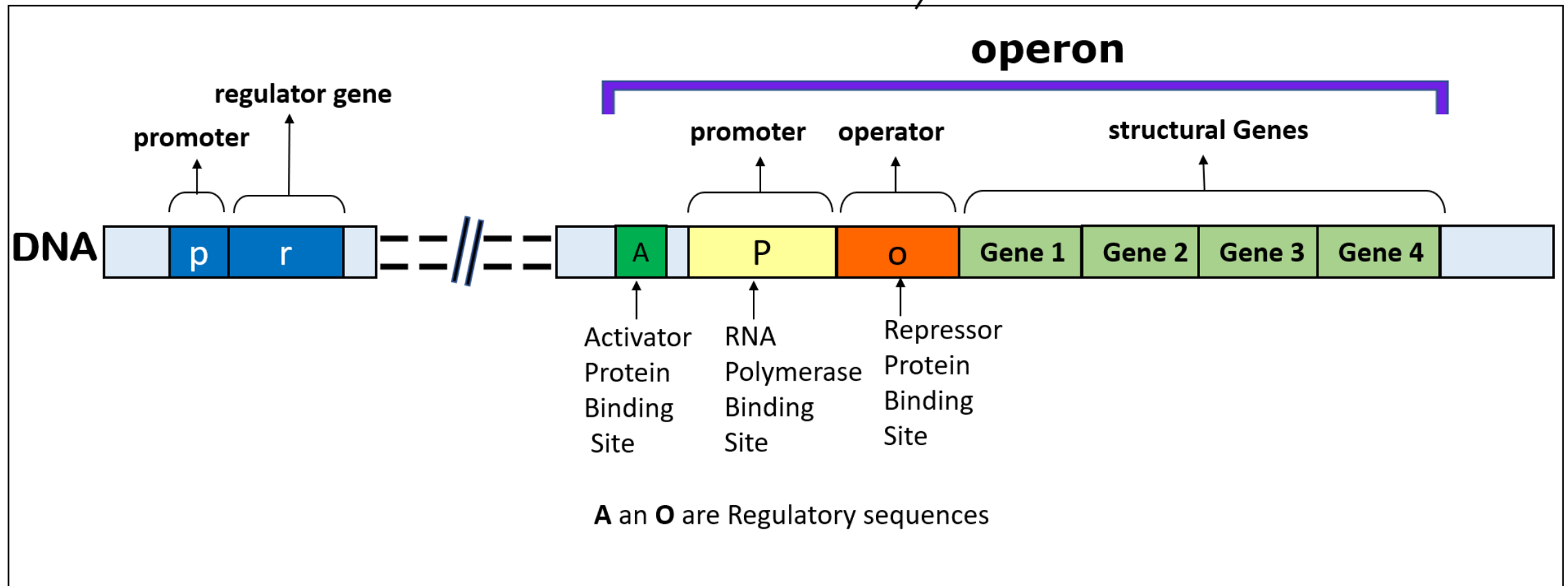
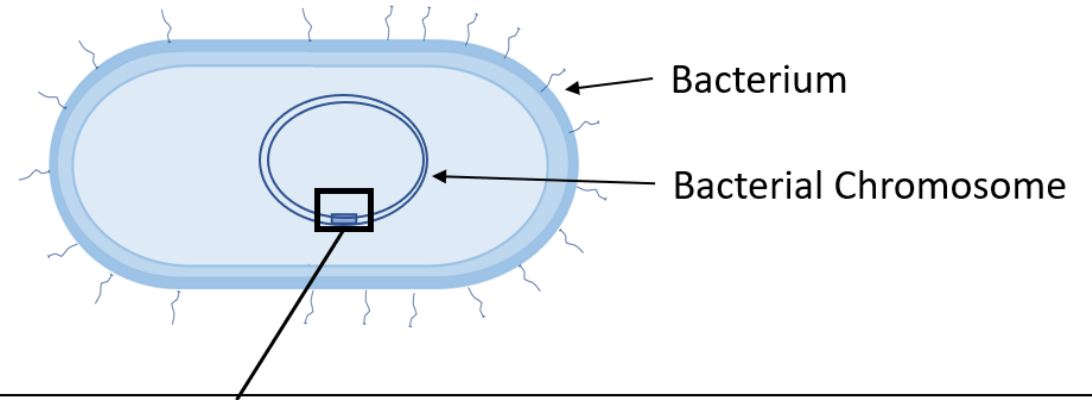
- **DNA:** A single, large DNA molecule (e.g., *E. coli* has a chromosome of about 4.6 million base pairs).
- **Supercoiling:** Enzymes called **topoisomerases** are responsible for supercoiling.
 - **DNA Gyrase (a type of topoisomerase II):** Introduces negative supercoils, which helps in compacting the DNA and also in separating the strands for replication and transcription.
- **Associated Proteins:** Prokaryotic DNA is associated with some proteins, but they are not the histones found in eukaryotes. These proteins help organize the supercoiled loops of the DNA.



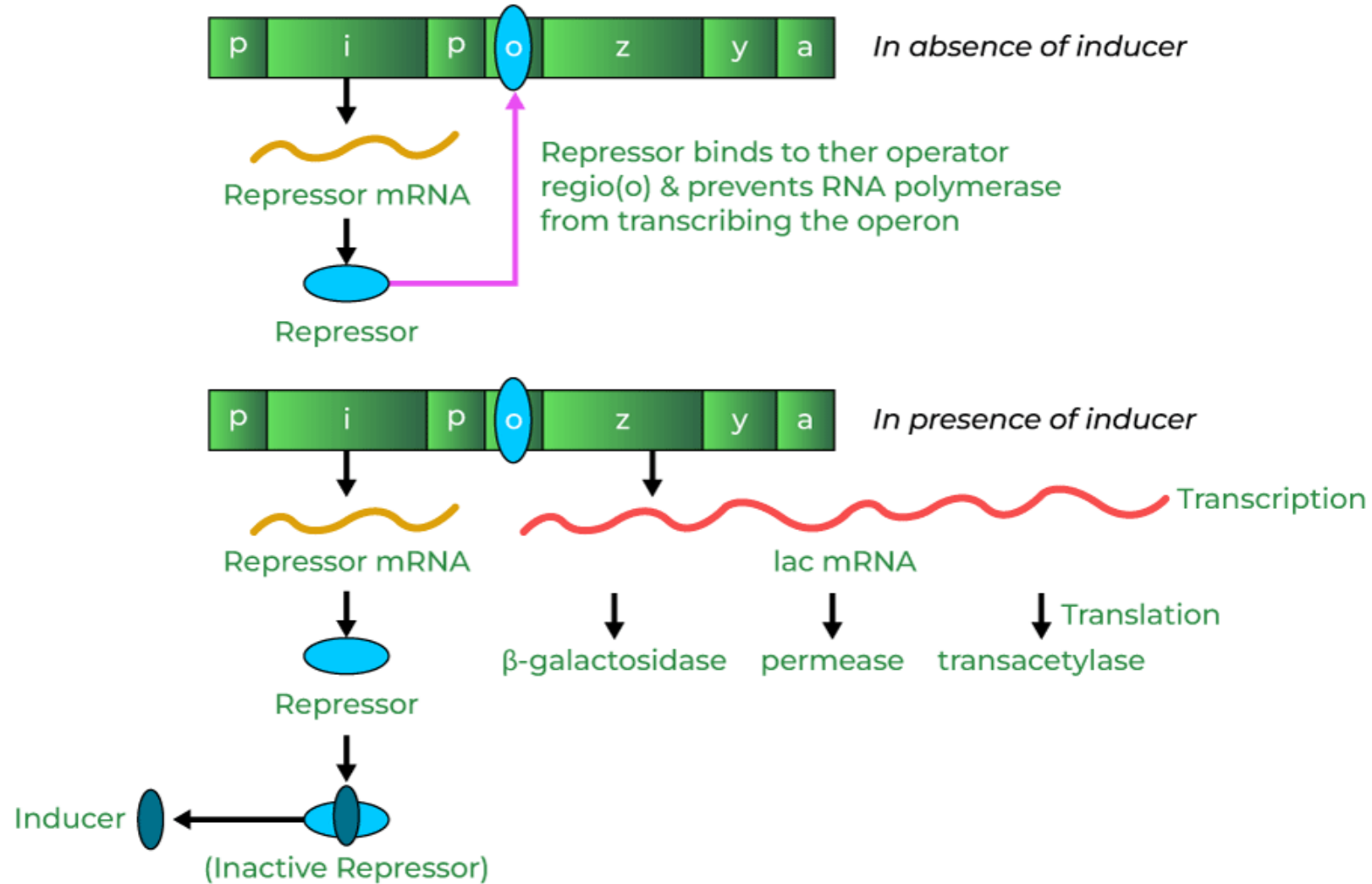
Gene Organization

- **High Gene Density:** A key feature of prokaryotic genomes is their efficiency. A very high percentage of the DNA sequence (often >90%) consists of coding genes.
- **Lack of Introns:** Genes are typically continuous, uninterrupted coding sequences.
- **Operons:** A powerful and important organizational unit.
 - An **operon** is a cluster of genes that are co-regulated and transcribed together as a single mRNA molecule from a single promoter.
 - The genes in an operon typically encode proteins that work together in a single metabolic pathway.
 - **Example:** The ***lac operon*** in *E. coli* contains genes for metabolizing lactose. All three genes are turned on or off together in response to the presence or absence of lactose.

Operon Structure



Lac Operon



Plasmids (Extrachromosomal DNA)

- **Definition:** Many prokaryotes also contain **plasmids**, which are small, circular DNA molecules that are separate from the main chromosome.
- **Properties:**
 - They replicate independently of the main chromosome.
 - They carry genes that are not essential for survival but provide a selective advantage (e.g., antibiotic resistance, genes for toxin production).
- **Significance:** Plasmids are a major mechanism for **horizontal gene transfer** in bacteria, allowing genetic information to be shared between organisms.

Eukaryotic Genome Organization

The eukaryotic genome is vastly more complex than its prokaryotic counterpart, with much larger size, multiple chromosomes, and elaborate packaging.

General Characteristics:

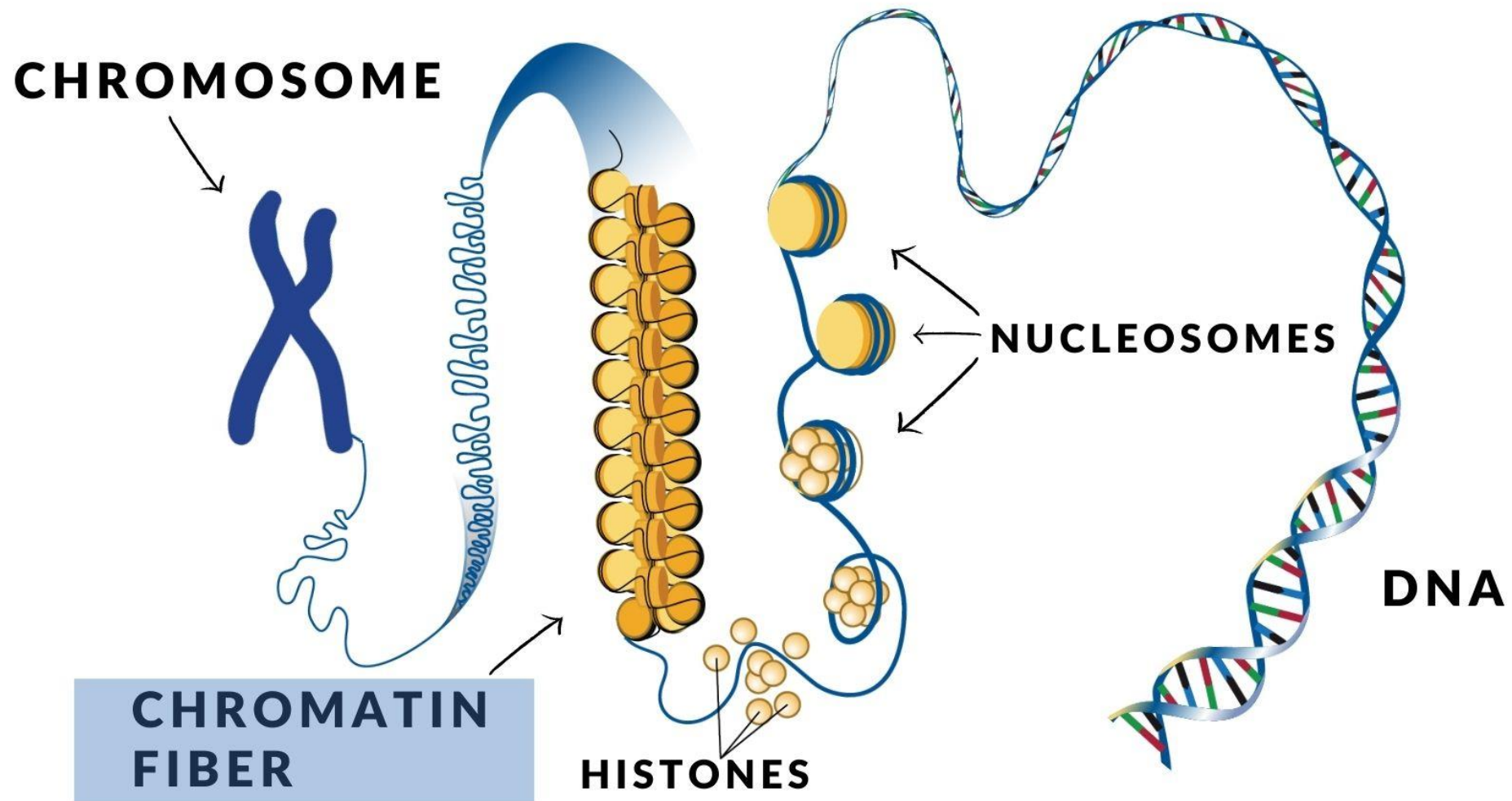
- **Multiple, Linear Chromosomes:** The genome is divided into several linear chromosomes, each consisting of a single, long DNA molecule.
- **Location:** The chromosomes are confined within a membrane-bound **nucleus**.
- **Chromatin:** The DNA is not naked; it is tightly associated with proteins, forming a complex called **chromatin**.

Chromatin: The DNA-Protein Complex

Histones: The primary proteins associated with DNA are called **histones**.

- Histones are small, positively charged proteins (rich in lysine and arginine).
- Their positive charge allows them to bind tightly to the negatively charged phosphate backbone of DNA.
- There are five major classes of histones: H1, H2A, H2B, H3, and H4.

Chromatin



Chromatin: The DNA-Protein Complex

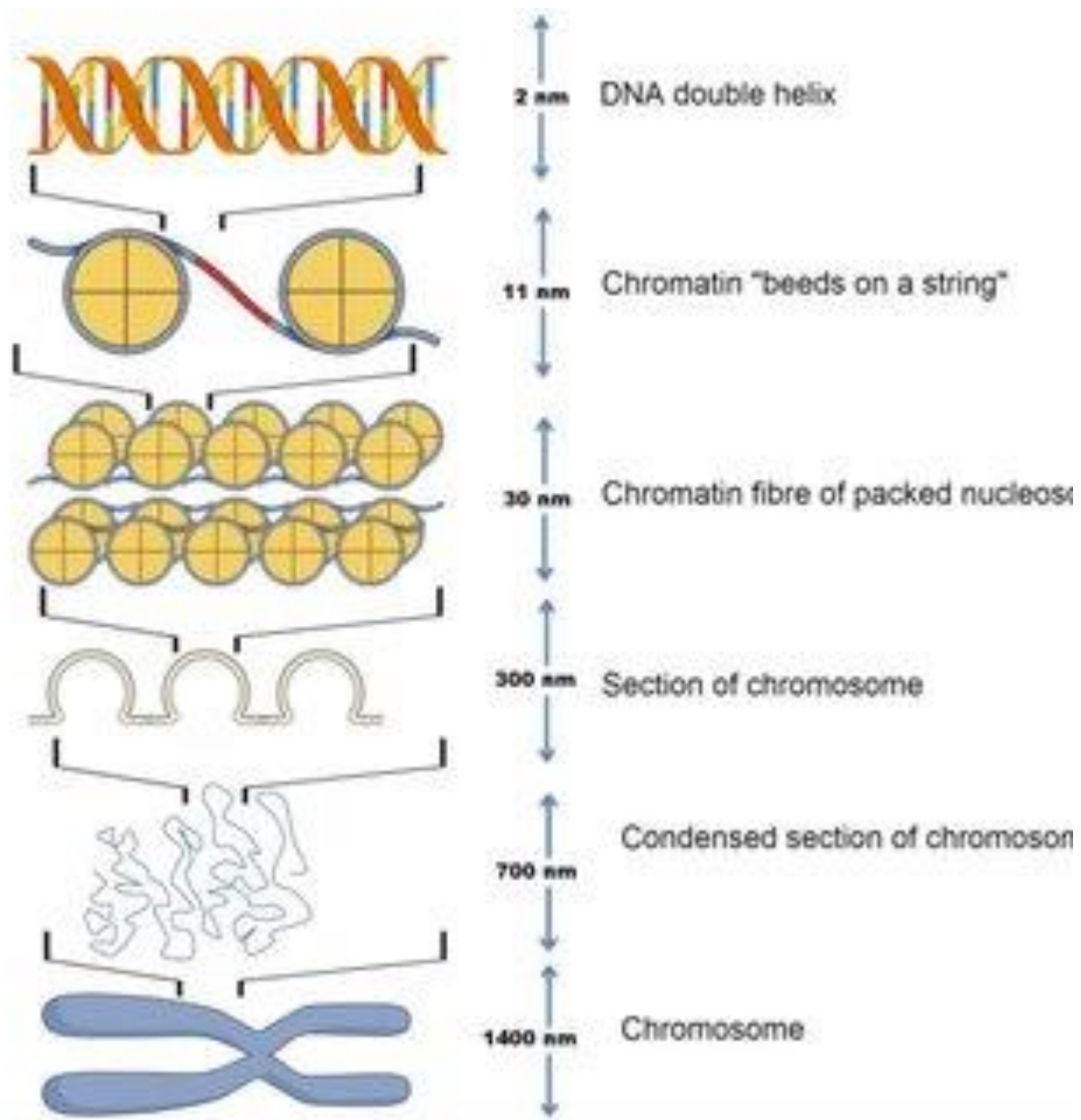
The Nucleosome: The Fundamental Unit:

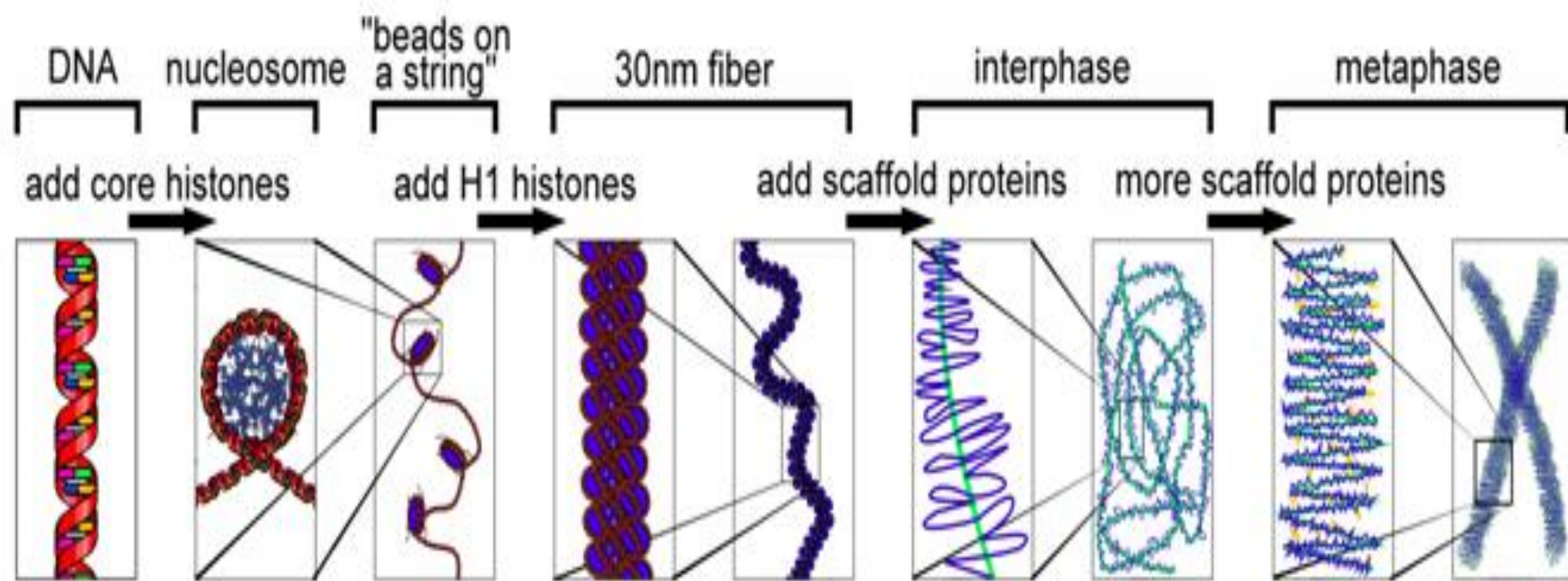
- The basic unit of chromatin structure is the **nucleosome**.
- It consists of a segment of DNA (about 147 base pairs) wrapped approximately 1.7 times around a core of eight histone proteins (an **octamer**), which contains two copies each of H2A, H2B, H3, and H4.
- This structure gives chromatin a "beads on a string" appearance.

Levels of Chromatin Compaction

The DNA in a human cell is over 2 meters long, yet it fits inside a nucleus only a few micrometers in diameter. This is achieved through multiple levels of compaction:

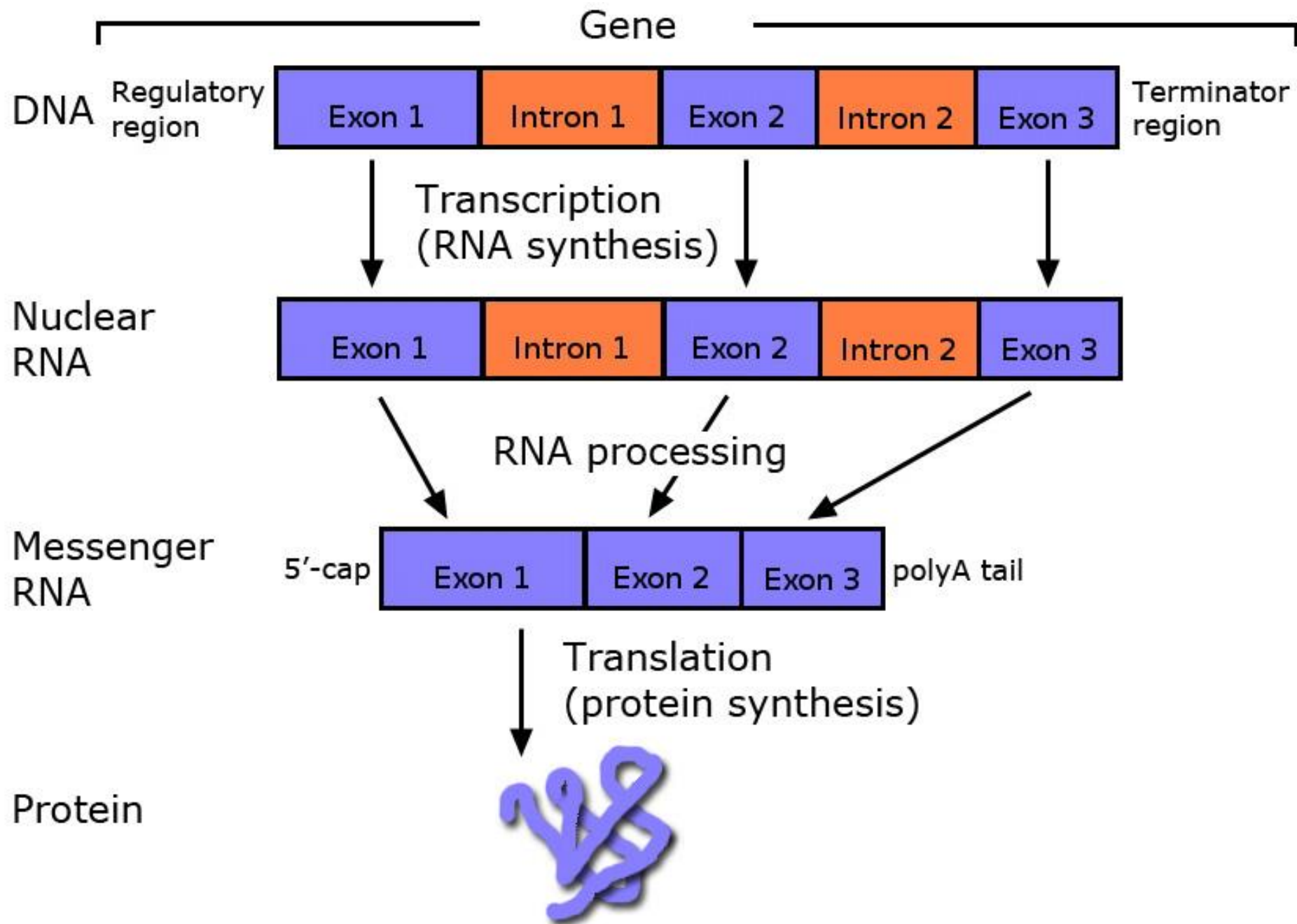
- **"Beads on a String" (11 nm fiber):** The basic nucleosome structure. The DNA is still relatively accessible.
- **30 nm Chromatin Fiber:** The "beads on a string" fiber is further coiled into a thicker, more compact structure, often stabilized by the H1 histone. This can take the form of a **solenoid** (a helix of nucleosomes) or a **zigzag** model.
- **Loops and Scaffolds:** The 30 nm fiber is organized into large loops (30-100 kbp), which are anchored to a non-histone protein **nuclear scaffold**.
- **Mitotic Chromosome:** During cell division, the looped domains become highly condensed, forming the visible, rod-like **mitotic chromosomes** we see under a microscope.





Gene Organization and Content

- **Low Gene Density:** A key difference from prokaryotes. Only a small fraction (around 1-2%) of the human genome codes for proteins.
- **Introns and Exons:** Eukaryotic genes are typically fragmented.
 - **Exons:** The coding regions that are eventually expressed.
 - **Introns:** The non-coding intervening sequences that are transcribed but later removed from the RNA transcript during a process called **RNA splicing**.



Gene Organization and Content

Extensive Non-Coding DNA: A large portion of the eukaryotic genome does not code for proteins but is still essential for genome function.

- **Regulatory Sequences:** Promoters, enhancers, and silencers control gene expression.
- **Repetitive DNA:** Regions of the genome with repeated sequences.
 - **Tandem Repeats:** Sequences that are repeated one after the other (e.g., centromeres, telomeres). **Minisatellites (VNTRs)** and **microsatellites (STRs)** are highly variable tandem repeats used in DNA fingerprinting.
 - **Interspersed Repeats:** Repetitive sequences scattered throughout the genome, often originating from ancient viruses called **retrotransposons** (e.g., SINEs like Alu elements, LINEs).

Chromosome Structure: Centromeres and Telomeres

- **Telomeres:**

- **Definition:** The repetitive DNA sequences (e.g., TTAGGG in humans) and associated proteins at the ends of linear chromosomes.
- **Function:** They protect the ends of the chromosome from degradation and fusion with other chromosomes. They also solve the "**end replication problem**," where DNA polymerase cannot replicate the very end of a linear chromosome. The enzyme **telomerase** adds these repeats to the ends.

- **Centromeres:**

- **Definition:** A specialized region of the chromosome that appears as a constriction. It consists of highly repetitive DNA sequences.
- **Function:** Essential for proper chromosome segregation during mitosis and meiosis. The **kinetochore**, a protein complex, assembles on the centromere and serves as the attachment site for spindle fibers.

THANK YOU